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Unlocking the power of the topic content in news headlines: BERTopic for predicting Chinese corporate bond defaults

Wenjin Tang [a,b], Hui Bu [a,b], Yuan Zuo [a,b,*], Junjie Wu [a,b]

[a] School of Economics and Management, Beihang University, Beijing 100191, China
[b] Key Laboratory of Data Intelligence and Management (Beihang University), Ministry of Industry and Information Technology, Beijing 100191, China

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ABSTRACT

This study explores the thematic content of news headlines and assesses their predictive power for corporate bond defaults. It establishes a data-driven framework, that emphasizes transparency and interpretability through the incorporation of explainable AI (xAI) techniques. The interpretable AI method, BERTopic, is applied to analyze news headlines from Jan. 1, 2014, to Aug. 31, 2021. A total of 18 economically inrerpretable topic measures are derived by combining similar topics among 78 original topics, offering insights into bond issuers' operational behavior and associated risks. Integrating the BERTopic model, various machine learning prediction models, the SHapley Additive exPlanations (SHAP) approach, and feature combination approaches, this study uncovers the incremental information contributed by news headlines beyond financial ratios and economic variables. The inclusion of these topic measures significantly enhances the predictive performance of first-time corporate bond defaults within a 3-month horizon. Additionally, the robustness of news headlines' information value is validated by extending the sample and employing an alternative study sample with differing credit risk scenarios, diverse markets, and even distinct news sources.

1. Introduction

News media serves as a crucial information intermediary in capital markets (Peress, 2014), offering valuable insights into various aspects, including firms' future earnings (Tetlock et al., 2008), future stock returns (Fang and Peress, 2009), macroeconomic indicators (Larsen and Thorsrud, 2019), and accounting fraud (Dyck et al., 2010). Given the diminishing early warning capability of third-party credit ratings, particularly evident in the Chinese corporate bond market as shown in Table S.1, enhancing credit risk evaluation and default prediction has become a crucial concern. One promising approach to enhance prediction accuracy involves augmenting the information sets used in the analysis. Therefore, this study investigates the potential of news to improve the prediction performance of corporate bond defaults. Furthermore, this study aims to enhance the interpretability and transparency of the entire modeling process (Xu et al., 2021; Wang, 2021), ultimately providing reliable insights for regulatory authorities and investors.

Topic modeling offers a superior and effective method compared to frequency/count-based approaches for extracting contextual meaning and semantic relationships among words, preserving the rich informational structure of unstructured news data (Huang et al., 2020). It has been successfully applied to various long financial texts, including news articles (Larsen and Thorsrud, 2019), 10-K

* Corresponding author at: School of Economics and Management, Beihang University, Address: No. 37 Xueyuan Road, Beijing 100191, China.
E-mail address: zuoyuan@buaa.edu.cn (Y. Zuo).

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reports (Hoberg and Lewis, 2017; Brown et al., 2020), SEC comment letters (Lowry et al., 2020), analyst reports (Huang et al., 2018; Bellstam et al., 2021), and conference call transcripts (Huang et al., 2018) .[1] However, the performance of topic modeling, particularly the widely used LDA model Blei et al. (2003), in handling short texts has not been sufficiently explored. Thus, this study analyzes the thematic content of the news headlines by topic modelling algorithms.

This study employs the BERTopic model[2] to investigate the thematic content in news headlines for predicting corporate bond issuers' first-time defaults, aiming to mimic the reading behavior of investors and offer cost-effective ways for credit risk assessment. This study validates the effectiveness of the BERTopic model in identifying economically interpretable topics based on Chinese corporate bond issuers' news headlines spanning from Jan. 1, 2014, to Aug. 31, 2021. The BERTopic model effectively discerns 78 distinct topics, subsequently streamlined into 18 combined topics leveraging domain knowledge.[3] The derived topics facilitate the construction of measures, enabling a timely and comprehensive assessment of issuers' behavior and risk conditions.

In an effort to enhance early-warning capabilities for corporate bond defaults, these topic measures are incorporated into machine learning methods alongside financial ratios and economic variables. The employed models include XGBoost, SVM, and ANN. Our findings demonstrate that models incorporating the topic measures exhibit additional predictive power for a 3-month horizon, outperforming models relying solely on financial ratios and economic variables. The incremental information value of news headlines extracted by BERTopic is proved to be robust not only across various classification models but also diverse feature combination approaches, as well as different samples and datasets.[4] This study makes significant contributions to the areas of text analysis and credit risk evaluation, which is the first contribution of this paper. Because this revenue opens new possibilities for cost-effective credit risk assessment using news headlines.

The second contribution is the transparent data-driven framework and explainable AI (xAI) techniques adopted by this study. Our data-driven framework combines models like BERTopic, machine learning algorithms, the SHapley Additive exPlanations approach (SHAP, Lundberg and Lee, 2017), and feature combination methods. This enhances transparency when extracting thematic information from news headlines to predict bond defaults and gains insight into the inner workings of the black box machine learning models. For example, we assess topic informative value through comparisons with machine learning models using different input variables, with or without BERTopic-derived topic measures. By leveraging the SHAP approach (Ariza-Garzón et al., 2020; Bussman et al., 2021), we deepen our understanding of the underlying mechanisms and the significance of topics in the machine learning models.[5]

2. Data and sample of corporate bond issuer

We analyze corporate bond default events in China from Jan. 1, 2014, to Aug. 31, 2021, sourced from the Wind Database. Our focus is on predicting the first-time default of an enterprise, which is more challenging.[6] We exclude first-time default bond issuers without complete financial data, resulting in a high-quality financial data sample of 65 issuers. To assess predictive value, we employe a time-controlled out-of-sample method, dividing the entire sample period into training and testing periods using the cutoff of Dec. 31, 2018. We train the model on 40 first-time defaulted issuers from Jan. 1, 2014, to Dec. 31, 2018, and evaluate it on 25 first-time defaulted issuers from Jan. 1, 2019, to Aug. 31, 2021.

[1] Larsen and Thorsrud (2019) shows that the topics extracted from news articles, has predictive power for key economic variables, and asset prices. Hoberg and Lewis (2017) and Brown et al. (2020) examines that the topics in 10-K reports can reflect corporate behavior and help detect financial misreporting. Lowry et al., (2020) investigates the regulator's role by identifying topics from SEC comment letters. Huang et al. (2018) compares the topic content of earnings calls, and subsequent analyst reports, to determine the analyst discovery and interpretation roles. Bellstam et al. (2021) uses topics in analyst reports, to measure innovation among firms.

[2] The BERTopic model, proposed by Grootendorst (2022), leverages the transformer-based model BERT (Devlin et al., 2019) to capture contextualized word embeddings and benefit from pre-trained language models, resulting in more accurate representations. This innovative approach overcomes the limitations of LDA, which requires sufficient word co-occurrence information to identify topics. Moreover, BERTopic's interpretability is invaluable, enabling us to comprehend how the algorithm identifies and categorizes thematic information within short texts. This transparency ensures the accuracy and reliability of extracted information and facilitates result explanation, empowering researchers in making more reliable and informed decisions based on extracted insights from news headlines.

[3] These combined topics encapsulate aspects related to financial performance, financing activities, intricate business transactions, business status and risks, management, shareholder behavior, and more.

[4] The entire modelling process is applied to an extended sample, addressing missing data issues, and an alternative study sample featuring distinct credit risk scenarios, varied market, and even different news source.

[5] Ariza-Garzón et al. (2020) applies the SHAP method to provide explainability to several ML models several well-known ML models in the context of P2P lending platforms' credit scoring. Bussman et al. (2021) confirms the usefulness of SHAP in increasing the transparency of complex ML models applied to loan decisions.

[6] Third-party credit ratings are already the reliable predictor of subsequently defaulted bonds, as they typically assign non-investment grade ratings to the bonds whose issuers have defaulted. The proportion of subsequently defaulted bonds for issuers in default but still in investment grade from 2014 to 2021 is 31.02% (76 out of 245), much smaller than that for defaulted issuers who have been downgraded to non-investment grade 82.41% (342 out of 415). The detailed statistics can be seen in Supplement 1 of Supplementary Material.

To address the severe imbalance between first-time defaulted and non-defaulted bond issuers,[7] we use a matching strategy[8] to construct a relatively balanced training sample using a matching ratio 1:3. Each first-time default bond issuer in the training sample is matched with 3 non-defaulted control issuers, considering factors like bond type, duration covering default time, not being previously sampled, and the availability of historical news data. This aims to mitigate selection bias due to potential heterogeneity in default levels across bond types and years. To examine whether the predictive power of the trained model is affected by the imbalance sample problem, we adopt several matching ratios (ranging from 1:1, 1:3, 1:5, to 1:10) for constructing the test samples. This approach offers insights into the model' s performance under conditions close real-world scenarios.

3. BERTopic on news headlines

3.1. News data collection

We download the news headlines data for our corporate bond issuer sample from the Choice database from Jan.1 2014 to Aug. 31, 2021. Using searches of the full/abbreviated/bond names of the enterprises, we match news to specific enterprises. A total of 15,956 news headlines are collected for 120 issuers in the training sample. Our emphasis on analyzing news headlines, rather than full articles, enables the efficient extraction of key information while effectively filtering out irrelevant details and noise.

3.2. Method and experimental analysis

We perform topic modeling using the BERTopic algorithm[9] considering its outstanding performance in short-text topic modeling and its interpretability. We employ the BERTopic algorithm on news headlines of issuers in the training sample, to avoid introducing information from the test sample and biasing the results. Using the Python package "BERTopic", we utilize a pre-trained BERT model designed for Chinese financial news by Li et al. (2018), aligning with our corpus. The BERTopic model is trained on pre-processed texts to enhance topic modeling effectiveness.[10] The algorithm initially identified 100 topics, and subsequently, based on hierarchical clustering results, merged them into 78 topics, as reported in Supplement 4.

3.3. Validation and interpretation of BERTopic outputs

To validate the BERTopic model outputs, we manually review high-frequency words and representative sentences in generated topics, assigning concise and intuitive labels to each topic,[11] following Brown et al. (2020). The effectiveness of BERTopic in identifying coherent, distinct, and economically meaningful topics in news headlines is demonstrated in Table S.3 of online supplementary material. Our results show BERTopic's efficacy in identifying general topics related to a firm's financial performance (e.g., "Income") and financing activities (e.g., "Debt issuance"). Moreover, BERTopic successfully detects complex business transactions like "Merger", business status topics such as "Production line", and management/shareholder behavior like "Shareholding pledge". BERTopic is also effective in revealing content related to business risks, including "Business difficulty and risk" and "Business affected by COVID-19″, along with industry-specific topics. For bond default prediction, we aggregate these topics by incorporating domain knowledge, terming them "combined topics". Specifically, we classify the 78 original topics into perspectives used by 10 external rating agencies

[7] As of August 2021, 65 issuers experienced their first-time bond default, while 25,772 corporate bonds from 3,337 enterprises, matured without default, spanning various types, such as listed and non-listed companies' bonds, short-term financing bills, medium-term notes, and private placements of corporate bonds. This results in a significant imbalance ratio of approximately 1:51 between first-time bond defaults and normally matured corporate bonds.

[8] Training a model directly on the raw severe imbalanced sample would result in a bias towards the majority class, hinder the identification of significant features, and therefore limit the model's ability to generalize when applied to new samples. Matching is a down-sampling strategy, which is a common approach used in finance and accounting literature to handle imbalance issues, such as predicting firm bankruptcy or distress in the credit risk area (Geng et al., 2015; Liang et al., 2016), and financial fraud (Dechow et al., 2011; Hoberg and Lewis, 2017; Dong et al., 2018).

[9] BERTopic works in four sequential steps. BERTopic initially utilizes the Sentence-BERT (SBERT) framework (Reimers and Gurevych, 2019) to generate document embeddings derived from any BERT-based sentence transformer model. BERT (Devlin et al., 2019) is a state-of-the-art transformer-based machine learning technology in natural language processing tasks, which considers the contextual representation of words. After that, BERTopic leverages UMAP (McInnes et al., 2018) to reduce the dimension of embeddings and uses HDBSCAN (McInnes et al. 2017) to cluster reduced embeddings. Lastly, a new class-based variation of TF-IDF (Grootendorst, 2022) is developed to extract the topic representation for each generated topic. The details can be found in Supplement 3.

[10] The pre-processing steps including removing all non-Chinese characters (e.g., punctuation and numbers) and bond issuer names, segmenting words and remove stop words functional words. The remaining words after space linking, are used as inputs for the BERTopic model to meet the requirement of the Python package "BERTopic" that the inout should be the English-style format. We also experiment by employing the complete text without any pre-processing as input to the BERTopic model, and find that the accuracy and reliability of the topic modeling results would be affected as shown in Supplement 5. We emphasize the importance of text pre-processing in domain-specific corpora, consistent with Egger and Yu (2022).

[11] We present the top-10 frequency words for all 78 topics in Supplement 4.

and the 6 Cs of credit, resulting in 18 combined topics, as shown in Table 1. Empirical evidence in Supplement 6 supports BERTopic as a superior choice compared to LDA when applied to news headlines.

4. Predictive power of topic measures for corporate bond defaults

4.1. Topic measures

We quantify the semantic content of enterprise-specific news based on BERTopic outputs. We adopt BERTopic to assign probabilities for each negative news to different topics and then combined topics. Similar to Tetlock et al. (2008), We construct measures in daily frequency, to ensure timeliness. To ensure an adequate volume of news, we use a historical period until day t to calculate the topic measures. Our focus is solely on the thematic content of negative news, using the dictionary developed by Bu et al. (2022).[12] The topic measures calculate the proportion devoted to each combined topic for firm-level news headlines over a historical period of until day t,[13] as shown in Eq. (1).

$$TOPIC_{j,i,t} = \frac{\sum_{n_t \in D(t)} P\left(NEGNEWS_{i,n,j}\right)}{\sum_j \sum_{n_t \in D(t)} P\left(NEGNEWS_{i,n,j}\right)}, \tag{1}$$

where $P(NEGNEWS_{i,n,\ j})$ represents the probability of news headline n of issuer i released on a day within a historical period of $D(t)$ until day t to the combined topic j. Given the quarterly frequency of financial ratios and economic variables, we set the historical period $D(t)$ here as 3 months. These topic measures allow for timely and comprehensive quantification on enterprises' operational behavior and risks. Besides, we can directly examine their usefulness for predicting corporate bond defaults. The collection of all 18 topic measures is defined as $X_{Topic,i,t} = \{TOPIC_{j,i,t}, j = 1, \ldots, 18\}$.

4.2. Out-of-sample predictive performance of topic measures

After constructing topic measures, we investigate information value of these topic measures for predicting first-time corporate bond defaults. We examine whether incorporating the constructed topic measures into a machine learning model including only financial ratios and economic variables can improve the out-of-sample predictive performance. The machine learning algorithms we employ include XGBoost, SVM, and ANN, given their ability in capturing non-linear and complex relationships among a lot of variables. The 5-fold cross-validation technique[14] is employed in the training sample, to evaluate the performance of a model more reliably and to select the best model among a set of candidate models. We establish the benchmark model that includes 16 financial ratios and 3 economic variables,[15] X_{FinEco}. We then introduce all 18 topic measures (X_{Topic}). Here, when different sources of data are used together as inputs, they are combined through a direct concatenation method.[16] The forecast horizon of 3-month is chosen. All input variables are constructed based on the available information three months before prediction. We use standard evaluation metrics including the area under the ROC curve (AUC), and recall to quantitatively assess the prediction performance.

We find several results, according to Table 2. Firstly, the incorporation of the complete set of topic measures (X_{Topic}) into the standalone X_{FinEco} model consistently improves performance metrics AUC and recall in both the 5-fold cross-validation tests of the training sample and the test sample, irrespective of the classifier used. This highlights the incremental information value of topic

[12] Our adopted specialized negative dictionary is developed by Bu et al. (2022) to identify credit risk in the Chinese bond market, consisting of 323 negative words. About 31.50% of the news headlines in our training sample are classified as negative, and our focus on negative news stems from the observation that, on average, defaulted issuers are dominated by negative news during the 11 months leading up to 1 month prior to default, suggesting a stronger association of negative news with credit risk, particularly in the early warning period. The details can be seen in Supplement 7.

[13] The probability that a text belongs to a combined topic, is defined as the highest probability of the text belonging to the original topics under the combined topic. The construction of the topic measures involves dividing the sum of all topics, following Brown et al. (2020). The values of the topic measures are normalized to a range of 0 to 1, which enables a standardized scale for comparison. Normalization is also useful in mitigating the influence of the number of negative news about different companies on the results. This is because our focus lies more on the distribution of topics rather than the absolute count of negative news.

[14] In the k-fold cross-validation, dataset D is randomly split into k mutually exclusive subsets (folds), $D_1, D_2, \cdots, D_k$, of approximately equal size of first-time defaulted issuer samples. Of the k subsamples, a single subsample is retained as a validation set for testing the model, and the remaining $k - 1$ subsamples are employed as the training set. The cross-validation process is then repeated$k$ times, with each k subsample employed exactly once as the validation set. The algorithm hyperparameters are selected based on AUC and recall.

[15] The specific definitions of financial ratios and economic variables are reported in Appendix table A.1.

[16] Besides the direct combination, four feature combination approaches are employed, including principal component analysis (PCA), kernel PCA with the Gaussian kernel, and generative autoencoder model, and economic orthogonal methods following Baker and Wurgler (2006). The results presented in online Supplement 8 consistently demonstrate that the inclusion of topic measures enhances predictive performance compared to models relying only on financial or economic variables, regardless of the variable combination approach employed.

Table 1
18 domain knowledge-driven topics, 78 news data-driven topics, and their corresponding relations.

ID	Domain knowledge-driven topics	News data-driven topics
1	Profitability	Changes in net profit relative to prior periods; General profitability performance; Income; Loss
2	Asset quality	Asset quality
3	Disclosure	Shareholding meeting; Periodic report
4	Credit Rating	Credit Rating
5	Market situation	Outperformance in stock price; Underperformance in stock price; Unusual price fluctuation; Market value; Maker and sector trends
6	Market trading and sentiment	Fund flow; Trading volume rankings; Block Trade; Trading activities in various brokerage branches; Institutional site visit; Number of shareholders; Index component
7	Financing	Debt issuance; Repurchase; Secondary offering; CSRC's review on stock issuance; Exchanges' review on bond issuance
8	Debt crisis	Debt crisis
9	Reorganization	Reorganization
10	Production and operation	Business difficulty and risk; Production capacity; Sales agreement signing; Guarantee; Project winning; Suppliers and customers; Business affected by COVID-19; Business condition; Clarification of cooperation relationship; Business cooperation; Business layout; Charitable donation; Project advancement; Government grants; Dividend payout
11	Personnel change	Personnel change
12	Asset proposal	Equity transfer; Asset replacement
13	Development strategy	Strategic cooperation; Subsidiary establishment; Merger; Corporate spin-off; Employee incentive; Acquisition results; Strategic investors; Changes in ownership and management; Equity investment; Acquisition procedure and details
14	Shareholding pledge and change	Changes in shareholding; Shareholding pledge
15	Legal risk	Judicial freeze; Judicial litigation
16	Regulatory inquiry and warning	Regulatory warning; Exchange inquiry
17	Violations of laws and rules	Illegal violation; Penalty
18	Industry-specific topics	Pharmaceutical R&D; Car sales; Wind power; Energy extraction; Lithium battery; Clarification on industry-related business; Land bidding; Solar power generation; New car launch; Electric bus; Marine transport; Urban development; Film, TV and live streaming; Electric vehicles; Car price adjustment

Table 2
The out-of-sample predictive performance of topic measures on corporate bond defaults.

	The 5-fold cross-validation tests for the training sample		The test sample with different matching ratio k							
			$k = 10$		$k = 5$		$k = 3$		$k = 1$	
	AUC	r	AUC	r	AUC	r	AUC	r	AUC	r
Panel A: XGBoost results										
X_{FinEco}	0.60± 0.11	0.62± 0.11	0.66	0.64	0.67	0.64	0.67	0.64	0.68	0.64
X_{FinEco}+X_{Topic}	0.73± 0.11	0.65± 0.14	0.85	0.76	0.82	0.76	0.79	0.76	0.80	0.76
Panel B: SVM results										
X_{FinEco}	0.62± 0.07	0.50± 0.16	0.56	0.56	0.54	0.56	0.51	0.56	0.44	0.56
X_{FinEco}+X_{Topic}	0.66± 0.09	0.70± 0.13	0.80	0.80	0.77	0.80	0.73	0.80	0.74	0.80
Panel C: ANN results										
X_{FinEco}	0.59± 0.06	0.35± 0.09	0.66	0.68	0.69	0.68	0.67	0.68	0.66	0.68
X_{FinEco}+X_{Topic}	0.67± 0.09	0.52± 0.17	0.81	0.76	0.77	0.76	0.71	0.76	0.70	0.76

Note: Under five-fold cross-validation test of the training sample, the mean and standard deviation of the evaluation metrics on the five validation sets are reported. The evaluation metrics for the test sample with different matching ratios are reported. The matching ratio k means that each first-time defaulted bond issuer is matched with k non-defaulted bond issuers to construct the test sample. "r" represent the prediction evaluation metrics recall. X_{FinEco} is the collection of 16 financial ratios that reflect profitability, liquidity, leverage level, solvency, and operating efficiency, and 3 quarterly economic variables. X_{Topic} is the collection of 18 topic measures that we construct in Section 4.1. The forecast horizon is 3 months.

content, demonstrating its robustness across various classifiers.[17] Secondly, SVM achieves the highest recall improvement of approximately 20% across different matching ratios in the test set, reaching a value of 0.80. If prioritizing AUC, the XGBoost model emerges as the optimal choice, achieving AUC values ranging from 0.80 to 0.85 on the test sample.

[17] For the bond-issuing enterprises with missing financial data, we fill the missing values using their previous values, thereby expanding the number of first-time defaulted issuers from 65 to 190. In Supplement 9, we present the out-of-sample prediction results of the five-fold cross-validation of the training set and the test set, using XGBoost, SVM and ANN classifiers. The main conclusion is robust on this extended sample. In addition, we compare the predictive power of our developed topic measures with the predictive power of negative news number, calculated using Bu et al. (2022)'s methodology in Supplement 7. This examination aims to discern the distribution of topics that contribute incremental informational value beyond news sentiment. Our findings reveal that the performance of the topic measures surpasses that achieved by the number of negative news, as evidenced by the recall metric on the test sample. This finding highlights the incremental informational value of news headlines' topic content beyond the scope of their sentiment.

Thirdly, our trained models exhibit strong model generalization capabilities, particularly XGBoost and ANN, performing even better on the training set than that on the test set.[18] Additionally, as the matching ratio increases, recall remains stable, and AUC maintains an acceptable level for the test sample. Interestingly, AUC increases with higher matching ratios for XGBoost and SVM, indicating the robustness of our models to imbalanced data and their potential applicability in real-world scenarios.

4.3. Interpret the machine learning model with the SHAP approach

The SHAP approach, introduced by Lundberg and Lee (2017), provides a method for interpreting black box machine learning (ML) models. We apply the SHAP approach[19] to interpret the best model in terms of AUC and recall, the XGBoost model with X_{FinEco} and X_{Topic} as inputs. This interpretive process can be extended to other ML models. Fig. 1 presents a summary plot of the Shapley values for the top 10 contributing features across all instances, which provides a global perspective on how these features influence the XGBoost output. By examining the descending order of importance, we find that risky topics play a significant role in driving the model's outputs. Media coverage on topics such as "Violations of laws and rules", "Regulatory inquiry and warning", "Debt risk", "Rating" and "Financing" significantly positively[20] predict issuers' first-time bond defaults occurring 3 months later. In summary, the SHAP approach effectively explains the importance of topic measures in the XGBoost classifier, improving the interpretability of the whole modelling process.

5. Robustness results

The incremental information value of news headlines extracted by BERTopic is proved to be robust across diverse feature combination approaches, as presented in online Supplement 9. This information value persists in an extended sample that addresses missing financial data by maintaining previous values, as indicated in online Supplement 10. Furthermore, the generalizability and applicability of our proposed modeling process, along with the information value of news headlines, are confirmed on an alternate study sample featuring distinct credit risk scenarios, varied markets, and different news sources. Empirical findings in online Supplement 12 highlight the valuable information contained in both Thomson Reuters' and Choice's news data, contributing to an improved prediction of financially distressing events for listed companies.

6. Conclusion

How to obtain useful information for credit risk evaluation and early warning is a key research question. Our study highlights the usefulness of topic modeling, specifically the interpretable AI model, BERTopic, in extracting qualitative information from news headlines. The identified 78 topics extracted from bond issuers' news headlines are further combined into 18 economically interpretable topics, capturing the content related to financial performance, financing activities, complex business transactions, business status and risks, and so on. This enables us to construct timely and comprehensive measures on enterprises' operation behavior and related risks.

We develop topic measures that quantify the thematic content by calculating the topic distribution of issuer-level news headlines. Our findings indicate that topic measures can serve as early-warning signals, improving the performance of predictions for corporate bond defaults under a 3-month horizon. The incremental information value derived from news headlines, beyond traditional financial ratios and economic variables, are validated across diverse machine learning models and various feature combination methods. The integration of BERTopic, ML models, the SHAP technique increases the transparency and interpretability of the entire modeling process, resulting in more comprehensive and reliable outcomes. Furthermore, the generalizability and applicability of our propose modelling process, along with the information value of news headlines, are confirmed on an extended sample and an alternate study sample featuring distinct credit risk scenario, varied market, and even different news source.

CRediT authorship contribution statement

Wenjin Tang: Conceptualization, Investigation, Software, Writing – original draft. **Hui Bu:** Conceptualization, Data curation, Funding acquisition, Writing – review & editing. **Yuan Zuo:** Funding acquisition, Methodology, Project administration, Software, Writing – review & editing. **Junjie Wu:** Methodology, Supervision, Validation, Writing – review & editing.

[18] While SVM models show a slight decrease in effectiveness, the decline remains minimal and acceptable.

[19] We adopt a conditional logistic regression (LR) model, following Campbell et al. (2008) to interpret which topics contribute to predict corporate bond defaults, and find the results are similar, as shown in Supplement 11.

[20] Features predominantly represented by red (blue) dots on the right-hand side of the X-axis exhibit a positive (negative) effect on the prediction output, as observed in "Violations of laws and rules" ("ICR").

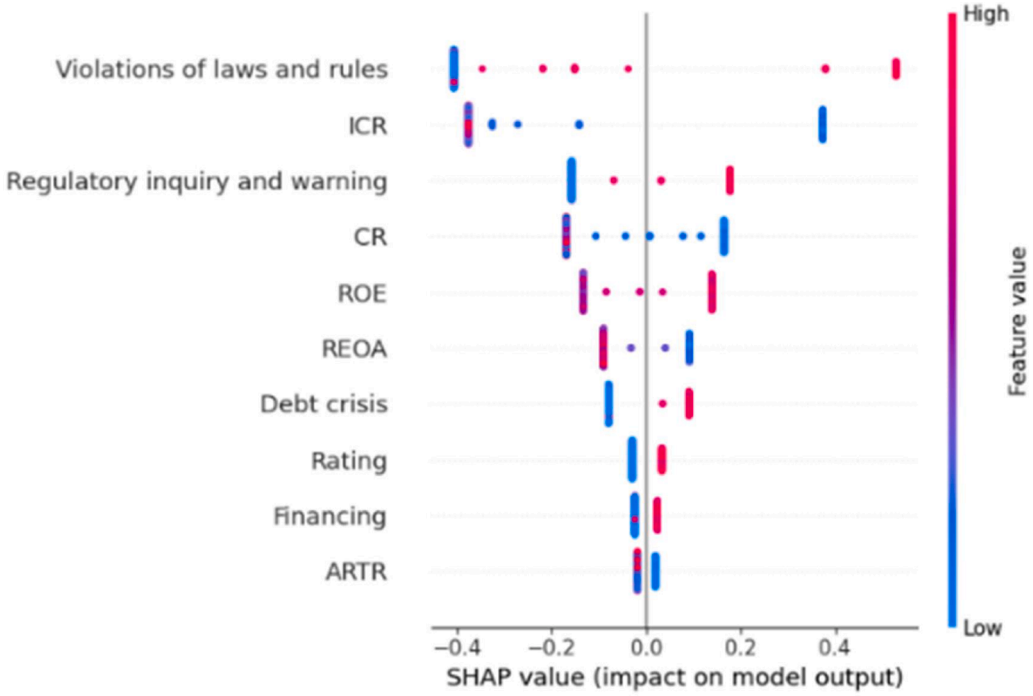

Fig. 1. SHAP summary plot of the XGBoost model for corporate bond defaults.
Note: This table shows the distribution of all instances' Shapley values for top-10 contributing features. On the Y-axis, features sorted according to their contributions are listed, measured through the sum of the absolute values of Shapley values of all instances. Each point in the graph represents the Shapley value a feature for a single case. The color indicates whether the feature is a high value (red) or a low value (blue). The X-axis of the graph represents the size and direction of the contribution of each feature to the prediction of classification.

Declaration of competing interest

We declare that we have no conflict of interest to this work.

Data availability

Data will be made available on request.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.frl.2024.105062.

Appendix

Table A.1
The definitions on financial ratios and economic variables used in prediction models.

Category	Variables
Financial ratios	Profitability: return on equity (*ROE*), return on total assets (*ROA*), retained earnings on total asset (*REOA*), net profit margin on sales (*NPMOS*)
	Liquidity: working capital ratio (*WCR*), quick ratio (*QR*), cash ratio (*CR*)
	Solvency: interest coverage ratio (*ICR*)
	Leverage: debt ratio (*DR*), debt to equity ratio (*DER*)
	Operating efficiency: total asset turnover ratio (*TATR*), accounts receivable turnover ratio (*ARTR*), inventory turnover ratio (*ITR*)
Growth of accounting data	growth rates of total assets (*GROA*), net profit (*GRONP*), and revenue (*GROR*)
Economic variables	year-on-year growth rates of quarterly PPI (PPI), fixed-asset investment (INV_FA), and aggregate financing to the real economy (AF_RE)

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